

Transport Across the Cell Membrane

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03_Transport_across_the_cell_membrane.pdf (26 slides) · Refs: Guyton & Hall 13th ed.; Ganong 25th ed.

Legend: ★ = high-yield · [added] = not on slides ·  = the lecturer's handwritten annotation. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

⚠ The lecturer **crossed out** Slide 9 (fluid-shift diagrams, marked "cancel") and items 2–6 of the Na^+/K^+ pump regulators (Slide 14). Both are shown here but marked as likely **not examinable**. Confirm with the lecturer.

 **Bridge from Lecture 02 (Cell Membrane):** Lecture 02 built the membrane and covered the first two diffusion types, **simple** (no carrier) and **facilitated** (carrier, saturable). This lecture finishes transport: **osmosis** (the third diffusion type), **active transport** (primary and secondary), and **vesicular transport**.

1. Title / Objectives

Objective	Likely tested as
★ Different methods of transport across the membrane	Osmole/osmolality definitions, tonicity of IV fluids, Na^+/K^+ pump facts, SGLT/GLUT, pino- vs phagocytosis
Types of intercellular communication	Only one conclusion line (gap junction, neuronal, endocrine); not taught on the slides

Flow: Osmosis (definition, osmotic pressure, osmole, osmolality vs osmolality, plasma 290, tonicity) → Active transport (definition, uniport/symport/antiport, primary Na^+/K^+ ATPase, secondary Na^+ -glucose) → Vesicular (endo-, exocytosis) → Conclusion → Case.

Transport across the cell membrane: one map

PASSIVE (with gradient, no ATP), small molecules			ACTIVE (against gradient)		VESICULAR (large)	
Simple	Facilitated	Osmosis	Primary	Secondary	Endocytosis	Exocytosis
No carrier • Lipid bilayer: O_2 , N_2 , $\text{CO}_2 \propto$ lipid solubility • Channels: leak or gated (voltage, mechanical, ligand, phospho) Linear with gradient, no maximum	Carrier protein e.g. glucose (GLUT) • Specific • Competition (glucose/galactose) • Saturates ($V_{\max}$) • More temp-sensitive	Water moves toward higher solute conc. Pressure depends on NUMBER of particles, not mass Plasma 290 mOsm/L = 5597 mmHg	ATP used directly Na^+/K^+ ATPase 3 Na^+ out 2 K^+ in ouabain / digoxin block	Uses the Na^+ gradient made by primary SGLT1/2: Na^+ + glucose in (symport) then glucose out via GLUT (facilitated)	IN Pinocytosis cell drinking: proteins, ECF Phagocytosis cell eating: bacteria, debris (some WBCs)	OUT Balance with endocytosis keeps area constant

Simple diffusion rate $\propto (\Delta C \times \text{surface area} \times \text{temperature}) \div (\text{distance} \times \sqrt{\text{molecular weight}})$

Carrier direction: uniport (1 substance, e.g. Ca^{2+} pump) · symport (same direction, Na^+ + glucose) · antiport (opposite, Na^+/K^+)

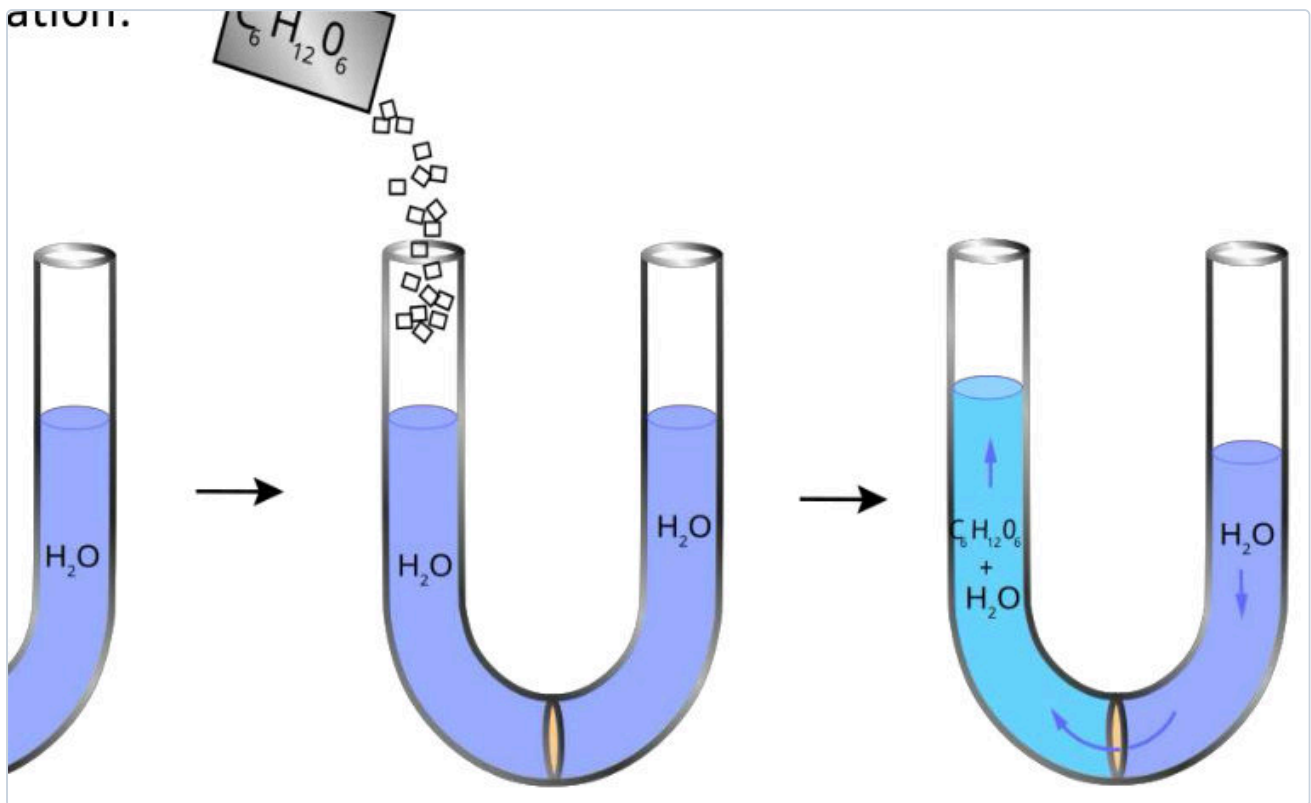
Active transport energy $\propto \log \Delta C$ · Na^+/K^+ pump uses 24% of cell energy, 70% in neurons, 90% in kidney

Added diagram: all transport methods on one map

2. Main Content (Lecture Order)

2.1 Osmosis (Slides 4-9)

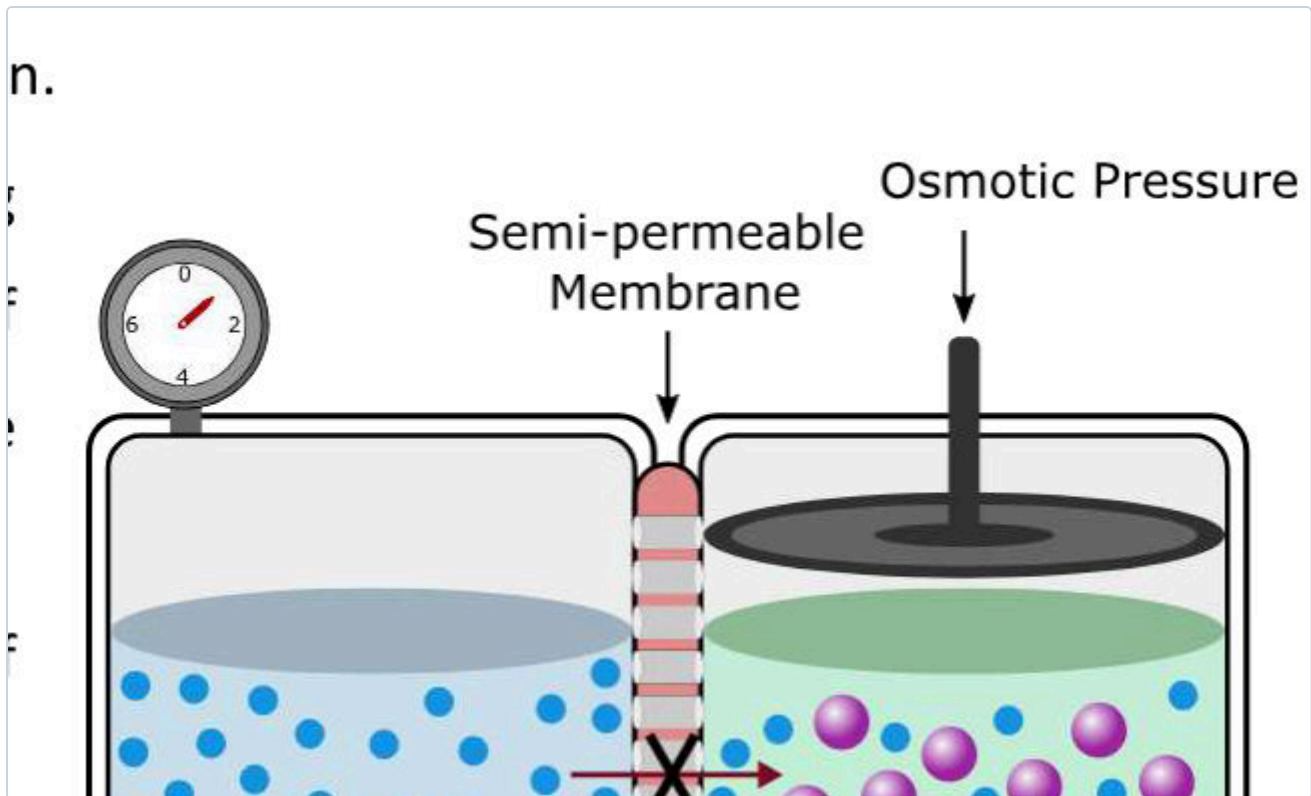
★ **Definition:** the tendency of **solvent (water)** to diffuse **down its own concentration gradient**, i.e., **toward the higher solute concentration**.



Slide 4: Water moves toward the side where solute is added

★ **Osmotic pressure** = the pressure that must be applied to the more concentrated solution to **prevent** osmosis.

- Measured in **mmHg**
- ★ Determined by the **NUMBER of particles per unit volume, not their mass**
- Osmosis $\propto$ number of particles / volume of solvent



Slide 5: Osmotic pressure opposes water flow across a semipermeable membrane

★★ Definitions (Slide 6):

Term	Meaning
Mole	Contains Avogadro's number = 6.022×10^{23} particles
★ Osmole	Number of osmotically active particles one mole liberates in solution
Glucose example	★ 1 mol glucose (180 g) = 1 osmole (doesn't dissociate)
NaCl example	★ 1 mol NaCl (58.5 g) = 2 osmoles ($\rightarrow \text{Na}^+ + \text{Cl}^-$)
★ Osmolality	Osmoles per kilogram of water
★ Osmolarity	Osmoles per litre of water
<i>Mnemonic [added]:</i> osmolaRity = per litRe; osmoLaLity = per kiLo (think "kg of water").	

★ Plasma osmolarity (Slide 7):

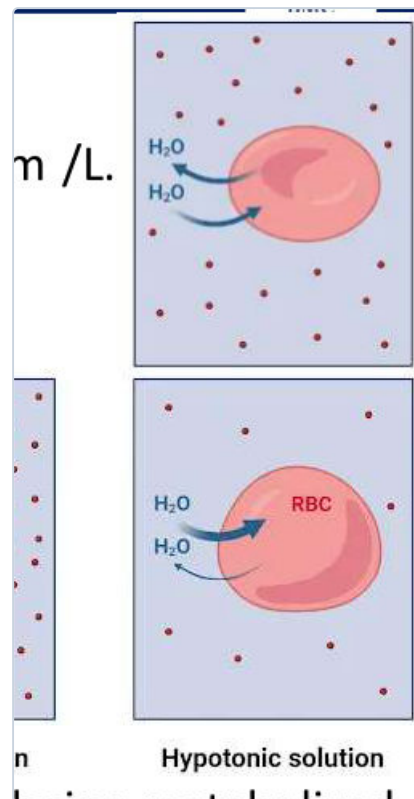
- Summing all ions would give **> 300 mOsm/L**, but body fluids are **not ideal solutions**: ionic interaction reduces the particles free to act
- ★ **Real plasma osmolarity = 290 mOsm/L $\approx$ 5597 mmHg osmotic pressure** (starred)
- ★ **ICF = plasma = ISF = 290 mOsm/L**

★★ Tonicity (Slide 8): osmolality of a solution **relative to plasma**

Type	Definition	Example / effect on RBC [RBC effect added]
★ Isotonic	Same as plasma (290 mOsm/L)	★ 0.9% NaCl or 5% glucose ; RBC unchanged
Hypertonic	Higher than plasma	RBC shrinks (crenates)
Hypotonic	Lower than plasma	RBC swells (may lyse)

★ **Key twist:** isosmotic solutions stay isotonic only if the solute doesn't enter cells or get metabolised.

- ★ **0.9% NaCl stays isotonic**
- ★ **5% glucose is isotonic at first, but once glucose is metabolised the net effect is a hypotonic (free-water) infusion**



Slide 8: RBCs in hypertonic vs hypotonic solutions

Slide 9 (✍ crossed out, "cancel"): fluid-shift (Darrow-Yannet) diagrams for diarrhoea, water deprivation, adrenal insufficiency, isotonic NaCl infusion, high NaCl intake, SIADH. The lecturer cancelled it, **but the closing case (Slide 24) uses exactly this diagram type**, so know the adrenal-insufficiency pattern (§7 Q1).

🔄 **Recap: Osmosis.** Water moves toward the higher solute concentration. Osmotic pressure depends on particle number, not mass. Glucose = 1 osmole/mol; NaCl = 2. Osmolality per kg, osmolarity per L. Plasma = 290 mOsm/L $\approx$ 5597 mmHg. 0.9% NaCl stays isotonic; 5% glucose becomes effectively hypotonic.

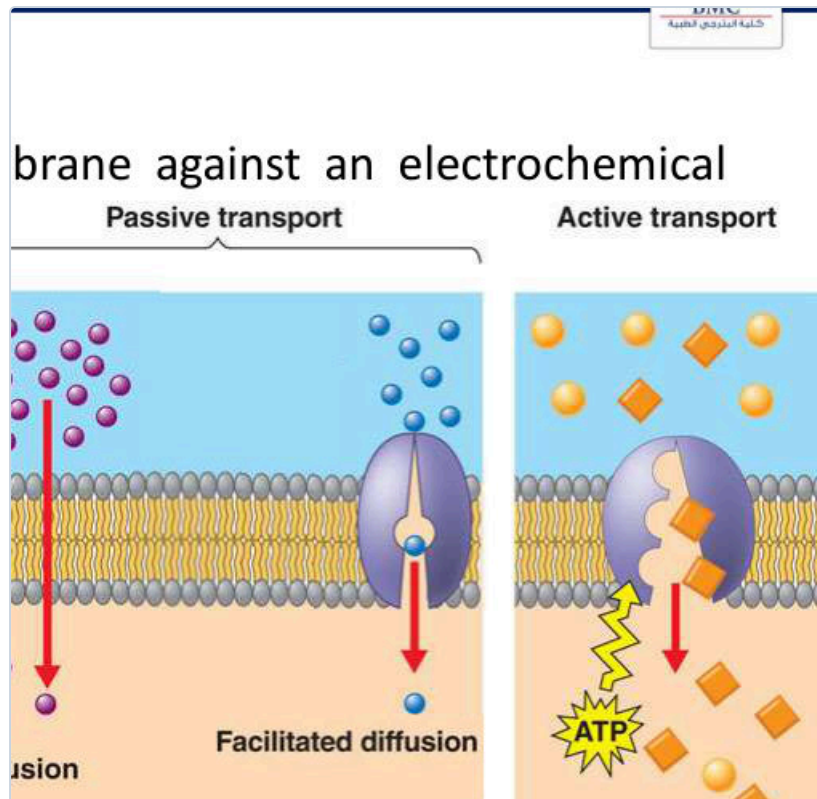
🔗 **Link:** Osmosis, like all diffusion, only goes downhill. To move substances against their gradient, the cell has to spend energy.

📺 **Up next: Active transport.**

2.2 Active Transport: General (Slides 10-11)

★ **Definition:** transport **against an electrochemical gradient**.

- Uses **specific carrier proteins**
- ★ **Needs energy**, from **ATP**; the carrier has **ATPase activity**
- ★ **Energy required $\propto \log \Delta C$**

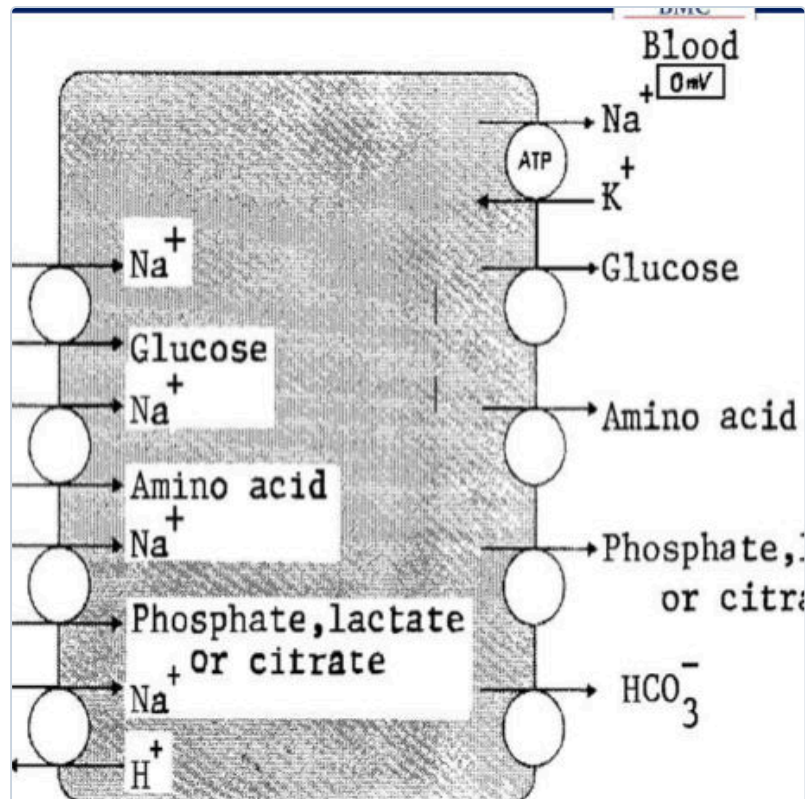


Slide 10: Passive (diffusion, facilitated) vs active transport

★★ Carrier direction types (Slide 11):

Type	Moves	Slide example
★ Uniport	One substance, one direction	Ca²⁺ pump
★ Symport (co-transport)	Two, same direction	Na⁺ + glucose from intestinal lumen into cells
★ Antiport (co-transport)	Two, opposite directions	Na⁺/K⁺ pump

Active transport is classified as **A. Primary** or **B. Secondary**.



Slide 11: Na^+ -coupled transporters (glucose, amino acids, phosphate)

2.3 Primary Active Transport: Na^+/K^+ ATPase (Slides 12-14)

★★ **Energy use:** 24% of cell energy; **70% in neurons**; **90% in kidneys** (90% highlighted).

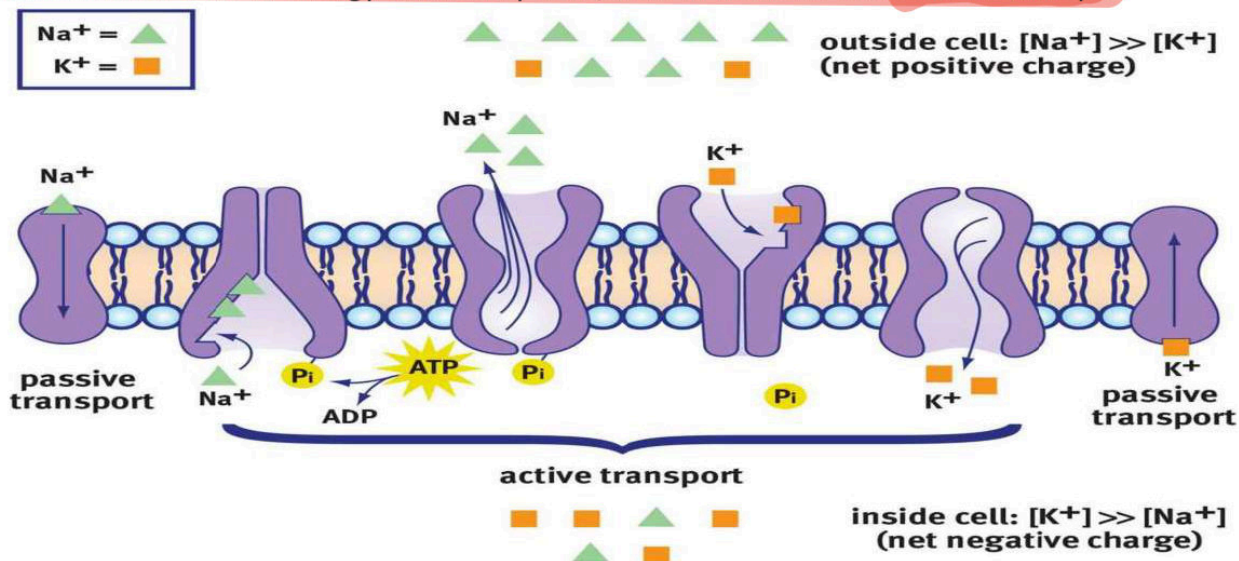
★★★★ **Key facts (Slide 13):**

Feature	Detail
★ Coupling ratio	3 Na^+ out : 2 K^+ in (3:2)
★ Inhibitors	Ouabain and related digitalis glycosides
Structure	Heterodimer ; separating the subunits abolishes activity
★ α subunit	Intracellular sites: 3 Na^+ , phosphorylation , ATP · Extracellular sites: 2 K^+ and ★ ouabain
β subunit	Anchors the complex in the membrane

[added consequence: the pump makes the inside more negative (electrogenic) and keeps Na^+ low inside, K^+ high inside: the ICF/ECF differences from Lecture 01.]

A. Primary (Na^+/K^+ ATPase pump).

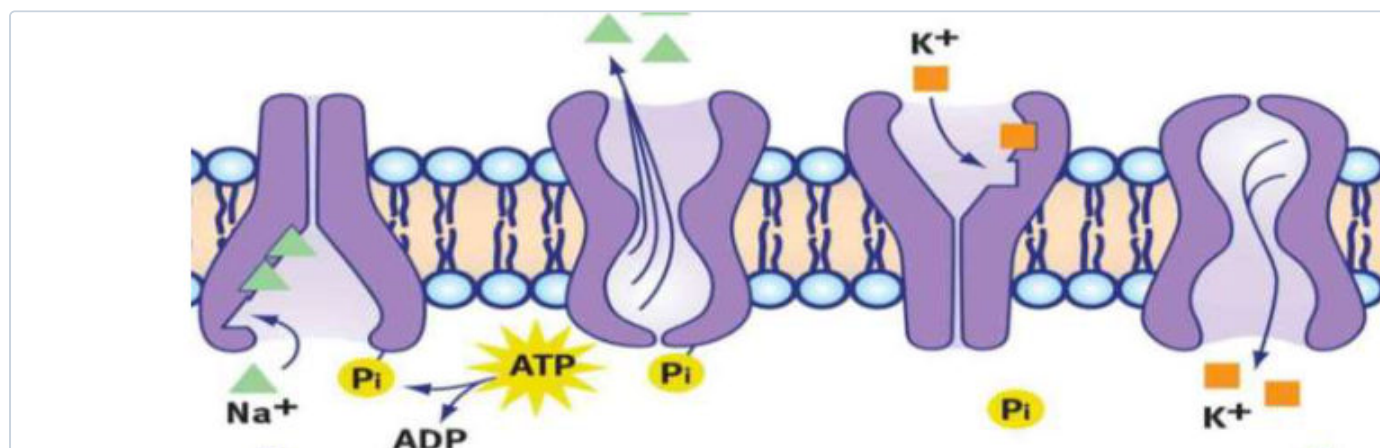
- Accounts for 24% of energy utilized by cells, 70% in neurons and 90% in kidneys.



Slide 12: Na^+/K^+ ATPase pump: 3 Na^+ out, 2 K^+ in

★ Pump cycle (Slide 14):

1. **3 Na^+ bind** (inside)
2. **ATP binds** → $\text{ATP} \rightarrow \text{ADP}$, phosphate transferred to the **phosphorylation site**
3. **Conformational change** → **Na^+ extruded** outside
4. **2 K^+ bind** (outside) → **dephosphorylation**
5. Return to the **original conformation** → **K^+ released** into the cytoplasm



Slide 13: Pump cycle stages

Regulation (Slide 14):

1. ★ **Substrate availability:** Na^+ , K^+ , **ATP** (the only item the lecturer kept)
2. ~~Thyroid hormones (genomic)~~ crossed out
3. ~~Aldosterone (secondary)~~ crossed out
4. ~~Insulin ($\uparrow$ activity by several mechanisms)~~ crossed out
5. ~~2nd messengers (cAMP, DAG)~~ crossed out
6. ~~Dopamine (phosphorylation-mediated inhibition)~~ crossed out

🔗 **Recap: Primary.** Na^+/K^+ ATPase is an antiport: 3 Na^+ out, 2 K^+ in, one ATP per cycle [added]. It uses 24% of cell energy (70% neurons, 90% kidney). α subunit holds the Na^+/ATP /phosphorylation sites (inside) and $\text{K}^+/\text{ouabain}$ sites (outside); β anchors. Blocked by ouabain/digitalis. Regulated mainly by Na^+ , K^+ , ATP availability.

🔗 **Link:** The Na^+ gradient the pump builds is stored energy. Secondary transport spends it to drag other molecules, like glucose, uphill.

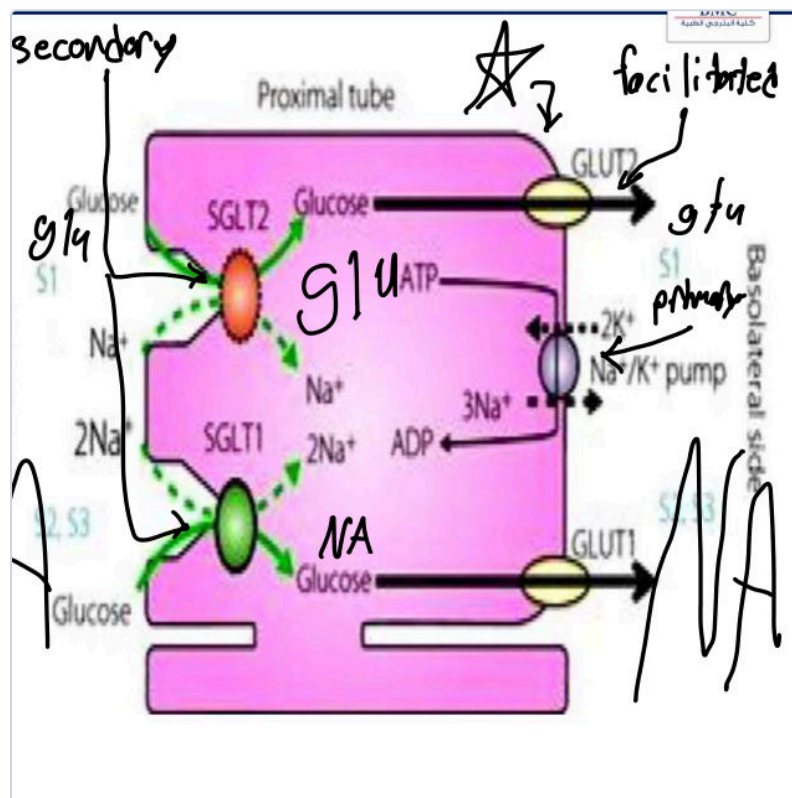
🔗 **Up next: Secondary active transport.**

2.4 Secondary Active Transport: Na^+ -Glucose (Slides 15-16)

★ **Definition:** carriage of glucose **secondary to the active transport of Na^+** . Energy comes from the **ionic concentration difference created by primary active transport** (not from ATP directly).

★★ **Steps (renal proximal tubule / gut):**

1. 📝 **Primary:** Na^+ pumped **out** at the **basolateral** membrane (Na^+/K^+ ATPase) → low intracellular Na^+
2. 📝 **Secondary:** at the **apical** membrane, **Na^+ + glucose bind the carrier (SGLT1 / SGLT2)** → shape change → **both enter** (symport)
3. This keeps intracellular Na^+ low, **maintaining the gradient**
4. 📝 **Facilitated:** **glucose leaves** down its gradient through **GLUT1 / GLUT2** (basolateral)



Slide 15: Proximal tubule: SGLT (secondary), Na^+/K^+ pump (primary), GLUT (facilitated), with the lecturer's labels

[added from the figure: SGLT2 carries 1 Na^+ per glucose (S1 segment); SGLT1 carries 2 Na^+ per glucose (S2/S3). SGLT2 inhibitors (dapagliflozin) are diabetes drugs; see the CVD risk lecture.]

🔁 **Recap: Secondary.** One epithelial cell uses three transport types: primary (Na^+/K^+ pump, basolateral), secondary (SGLT Na^+ -glucose symport, apical), facilitated (GLUT, basolateral).

🔗 **Link:** Diffusion and pumps handle small molecules. Proteins, bacteria and secretions are too big, so they travel inside vesicles.

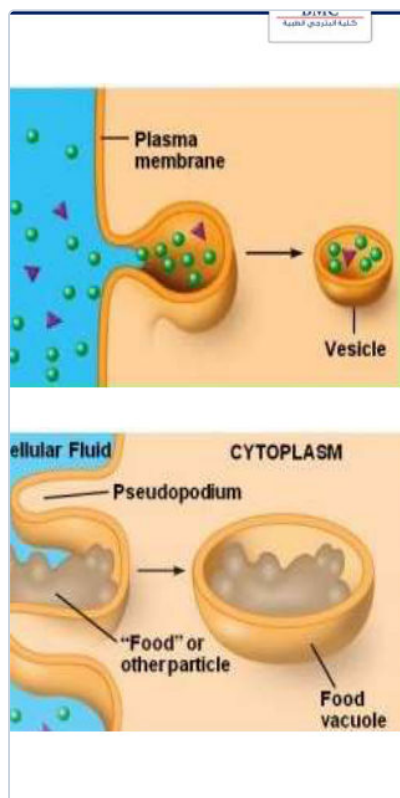
👉 **Up next: Vesicular transport.**

2.5 Vesicular Transport (Slides 17-19)

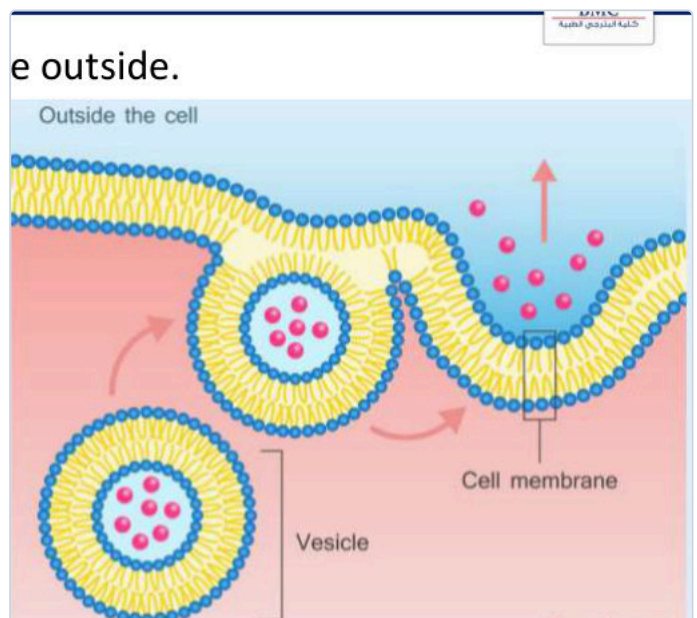
★ Movement of **macromolecules enclosed in membrane-bound vesicles**. Two types: **endocytosis** and **exocytosis**.

Type	Direction	Detail
★ Pinocytosis ("cell drinking")	In	The only way some very large water-soluble molecules (proteins) enter; small invaginations with ECF pinch off as pinocytic vesicles
★ Phagocytosis ("cell eating")	In	Same process for large particles: bacteria, dead tissue . ★ Only certain cells , e.g., some WBCs
★ Exocytosis	Out	Inside → outside

★ **Balance:** exo- and endocytosis balance, so membrane area **neither shrinks** (if endocytosis dominates) **nor becomes redundant** (if exocytosis dominates).



Slide 18: Pinocytosis and phagocytosis



Slide 19: Exocytosis

🔁 **Recap: Vesicular.** Pinocytosis (fluid, proteins), phagocytosis (bacteria, debris; only some WBCs), exocytosis (out). The two directions balance to keep membrane area constant.

🔗 **Link:** The conclusion ties together Lectures 01-03.

👉 **Up next:** Conclusion + case.

2.6 Conclusion (Slides 21-22)

- Water distributes between **ICF (40%)**, **ISF (15%)**, **plasma (5%)**
- Homeostasis = keeping the internal environment constant
- Membrane: amphipathic; proteins 55%, phospholipids 25%, cholesterol 13%, other lipids 4%, carbohydrates 3%
- Transport: **diffusion, active, vesicular**
- ★ **Intercellular communication: gap junctions, neuronal, or endocrine** (listed only here; the slides don't teach it further)

3. High-Yield Comparison Tables

Table A: All transport types

	Simple diffusion	Facilitated	Osmosis	Primary active	Secondary active	Vesicular
Gradient	Down	Down	Water down its gradient	Against	Against (for glucose)	n/a
Energy	None	None	None	ATP directly	Na⁺ gradient (ATP indirectly)	ATP [added]
Carrier	No	Yes	No (aquaporins help)	Yes (ATPase)	Yes (SGLT)	Vesicles
Saturable	No	Yes	No	Yes	Yes	n/a
Example	O ₂ , CO ₂	Glucose via GLUT	Water	Na ⁺ /K ⁺ pump, Ca ²⁺ pump	Na ⁺ -glucose (SGLT)	Proteins, bacteria

Table B: Uniport / symport / antiport (see §2.2)

Table C: IV fluids

Fluid	Initially	After distribution/metabolism
0.9% NaCl	Isotonic	Isotonic (stays in ECF) [ECF detail added]
5% glucose	Isotonic	★ Hypotonic (free water, distributes to all compartments)

Table D: Osmolality vs osmolarity

	Osmolality	Osmolarity
Per	kg water	L water
Plasma	≈ 290	≈ 290

4. Rules, Exceptions, Patterns

- **Osmotic pressure depends on particle NUMBER, not mass.**
- **NaCl gives 2 osmoles per mole; glucose gives 1.**
- **Real plasma osmolarity (290) < calculated (> 300)** because body fluids aren't ideal solutions.
- **5% glucose = isotonic in the bag, hypotonic in the body.**
- **Active = against gradient + carrier + energy ($\propto \log \Delta C$).**

- **Na⁺/K⁺ pump: 3 out, 2 in; ouabain/digitalis block the α subunit's extracellular site.**
- **Secondary active uses the Na⁺ gradient, not ATP directly.**
- **Glucose enters epithelia by secondary active (SGLT) and leaves by facilitated diffusion (GLUT).**
- **Phagocytosis only in certain cells;** pinocytosis is the only entry route for some large proteins.

5. Clinical Correlations

- **Digoxin toxicity** (digitalis glycoside) inhibits the Na⁺/K⁺ pump *[added: → ↑ intracellular Ca²⁺ via Na⁺/Ca²⁺ exchange → ↑ contractility; toxicity worsened by hypokalaemia]*.
- **5% dextrose given in excess** → free-water load → **hyponatraemia**, brain swelling *[added]*.
- **Oral rehydration solution** works through **Na⁺-glucose symport (SGLT1)**: water follows *[added]*.
- **SGLT2 inhibitors** (dapagliflozin) block renal glucose reabsorption → glycosuria; cardiorenal benefit (CVD lecture).
- **Addison's disease** (case): aldosterone deficiency → Na⁺ loss → hypovolaemic hyponatraemia → water shifts **into cells**.
- **Phagocyte defects** → recurrent bacterial infections *[added]*.

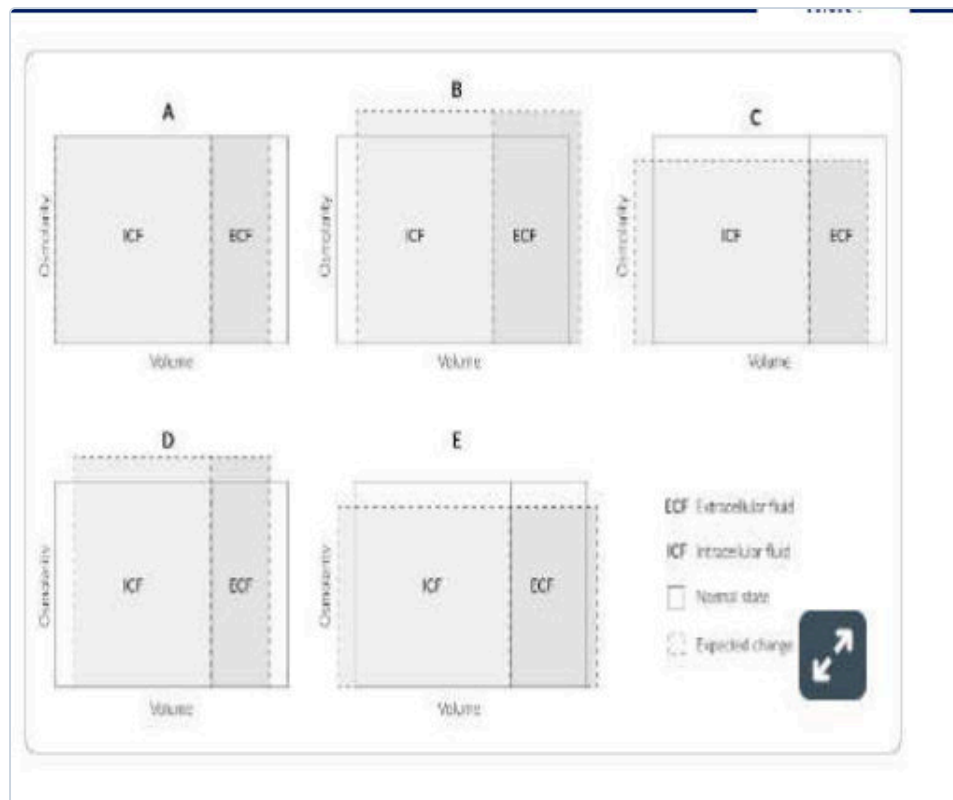
6. Rapid Revision (Last-Day)

- Osmosis: water → higher solute. Osmotic pressure (mmHg) ∝ **number of particles**.
- Avogadro **6.022 × 10²³**. **Glucose 180 g = 1 Osm; NaCl 58.5 g = 2 Osm.**
- **Osmolality = /kg; osmolarity = /L.**
- Plasma **290 mOsm/L = 5597 mmHg**; calculated > 300 (non-ideal solution).
- Tonicity = relative to plasma. **0.9% NaCl isotonic; 5% glucose isotonic → effectively hypotonic.**
- Active: against gradient, carrier, ATP (ATPase), **energy ∝ log ΔC.**
- **Uniport** (Ca²⁺ pump) · **symport** (Na⁺ + glucose) · **antiport** (Na⁺/K⁺).
- Na⁺/K⁺ ATPase: **24% energy (70% neurons, 90% kidney); 3 Na⁺ : 2 K⁺; ouabain/digitalis; heterodimer**; α = 3 Na⁺ + P + ATP inside, 2 K⁺ + ouabain outside; β anchors.
- Cycle: 3 Na⁺ bind → ATP → phosphorylation → Na⁺ out → 2 K⁺ bind → dephosphorylation → K⁺ in.
- Regulation: **substrate (Na⁺, K⁺, ATP)** (others crossed out by lecturer).
- Secondary: **SGLT1/2** (apical, Na⁺ + glucose in) → **GLUT1/2** (basolateral, glucose out); pump basolateral.
- Vesicular: **pinocytosis** (drinking, proteins) · **phagocytosis** (eating, bacteria, some WBCs) · **exocytosis**. Balance keeps membrane area constant.
- Intercellular communication: **gap junctions, neuronal, endocrine.**

7. Exam-Style Questions

Q1 (MCQ, lecture case, Slides 24-25). 32 ♀, 3 months of fatigue, anorexia, **10 kg weight loss**, ↓ libido; **type 1 diabetes**. Pulse 105, **BP 95/63**. **Skin and oral mucosa darkening**. Na⁺ **130**, K⁺ **5.2**, fasting glucose **74**. Which change in body-fluid volume and osmolarity is expected? **Answer: C** (the lecture's answer key, shown on Slide 25): **ECF volume ↓ and osmolarity ↓; water shifts from ECF to ICF → ICF volume ↑, ICF osmolarity ↓**. Reasoning (from the slide): **Addison disease** (adrenocortical insufficiency) → **aldosterone deficiency** → **↑ Na⁺ excretion** → loss of more solute

than water = **hyperosmotic fluid loss** → **hypovolaemic hyponatraemia**. Low ECF osmolarity pulls water into cells. Other causes of the same pattern: **loop diuretic overuse**, **21-hydroxylase deficiency**.



Slide 24: The case's answer diagrams (A-E)

Q2 (MCQ). Which IV fluid is isotonic in the bag but acts as hypotonic after infusion? a) 0.9% NaCl **b) 5% glucose** c) 3% NaCl d) Ringer's lactate **Answer: b.** Glucose is metabolised, leaving free water.

Q3 (MCQ). Ouabain binds the Na^+/K^+ ATPase at: a) β subunit, intracellular **b) α subunit, extracellular** c) α subunit, intracellular d) β subunit, extracellular **Answer: b.** The α subunit's extracellular sites bind 2 K^+ and ouabain.

Q4 (Calculation). How many osmoles does 1 mole of NaCl produce, and 1 mole of glucose? **Answer:** NaCl → **2** ($\text{Na}^+ + \text{Cl}^-$); glucose → **1**.

Q5 (Short answer). Explain how glucose is absorbed across a renal tubular cell using three transport types. **Answer: Primary active:** basolateral Na^+/K^+ ATPase keeps intracellular Na^+ low. **Secondary active:** apical SGLT1/2 uses the Na^+ gradient to bring $\text{Na}^+ + \text{glucose}$ in (symport). **Facilitated diffusion:** glucose exits basolaterally via GLUT1/2.

8. Exam Pitfalls & Traps

- Osmotic pressure depends on mass.** No: **number of particles**.
- Osmolality vs osmolarity:** per **kg** vs per **litre**.
- 5% glucose is always isotonic.** Only initially; it acts hypotonic once metabolised.
- Secondary active transport uses ATP directly.** No, it uses the **Na^+ gradient** from primary transport.
- Na^+/K^+ ratio: 3 Na^+ out, 2 K^+ in** (not 2:3, not 1:1).
- Na^+/K^+ pump is a symport.** It's an **antiport**; Na^+ -glucose is the symport.

7. **Glucose leaves the tubular cell by active transport.** It leaves by **facilitated diffusion (GLUT)**.
8. **All cells can phagocytose.** Only certain cells (some WBCs).
9. **⚠️ Crossed-out content:** Slide 9 fluid-shift diagrams ("cancel") and pump regulators 2-6 (thyroid, aldosterone, insulin, cAMP/DAG, dopamine). Likely not examinable, **but the closing case uses a fluid-shift diagram**. Ask the lecturer.
10. **⚠️ Slide typo:** "SGLUT-II or I" = **SGLT2 / SGLT1**; "oubain" = **ouabain**.
11. **Intercellular communication** is an objective but appears only in the conclusion (gap junction, neuronal, endocrine). Expect it in a later lecture.